Arpita Tripathi

B.Tech Information Technology Student | AI/ML & GenAI
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GitHub |Linkedin

EDUCATION

-Kamla Nehru Institute of Technology, Sultanpur	**2024-2028**
Bachelor of Technology (B-Tech) in Information Technology	8.4 CGPA

TECHNICAL SKILLS and INTERESTS

Languages: C++, Python, SQL
Generative AI: LangChain, RAG, LCEL, Hugging Face, Gemini API, OpenAI API
Machine Learning: TensorFlow, Keras, Scikit-learn, XGBoost, Transfer Learning
Web & Deployment: FastAPI, Streamlit, Docker, REST APIs
Data & Web Scraping: Pandas, NumPy, Matplotlib, Seaborn, BeautifulSoup, Tavily
Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, Google Colab

PROJECTS:

Multi-Agent Research System | LangChain, RAG, Gemini API, Tavily, BeautifulSoup, Streamlit

GitHub | **Live Demo**

- Built a multi-agent research system using LangChain and Gemini API for automated web research, content extraction, and information synthesis.
- Integrated Tavily Search API and BeautifulSoup for webpage discovery, scraping, and query-specific content extraction using LCEL/Runnables.
- Developed and deployed an interactive Streamlit application with secure API-key configuration for real-time, multi-source research.

Plant Disease Detection System | Python, TensorFlow, EfficientNet, Transfer Learning, Docker, Streamlit

GitHub | **Live Demo**

- Developed a deep learning-based plant disease classifier using EfficientNet transfer learning, trained on 20K+ PlantVillage images across 14 disease categories, with preprocessing and data augmentation.
- Achieved 96% accuracy through transfer learning and fine-tuning, and integrated the model into a Streamlit application for real-time disease prediction.
- Dockerized and deployed the application on Render, providing a scalable web interface for image-based disease detection.

Movie Recommendation System | Python, Pandas, Scikit-learn, Render, Docker

GitHub | **Live Demo**

- Developed a content-based movie recommendation system using cosine similarity on movie metadata to generate personalized recommendations.
- Processed and transformed movie features using Pandas and Scikit-learn, generating similarity scores to retrieve the most relevant movies based on user selection.
- Built a FastAPI REST API , Dockerized the application, and deployed it on Render for real-time movie recommendations.

Achievements:

- **Adobe University Hackathon 2026:** Qualified for Round 1.
- **Smart India Hackathon (SIH) 2025:** Qualified for the Semifinal Round.
- **Competitive Programming:** Solved **500+ DSA problems** across LeetCode, CodeChef, and GeeksforGeeks.
- **JEE Main 2024:** Achieved **95 percentile**.